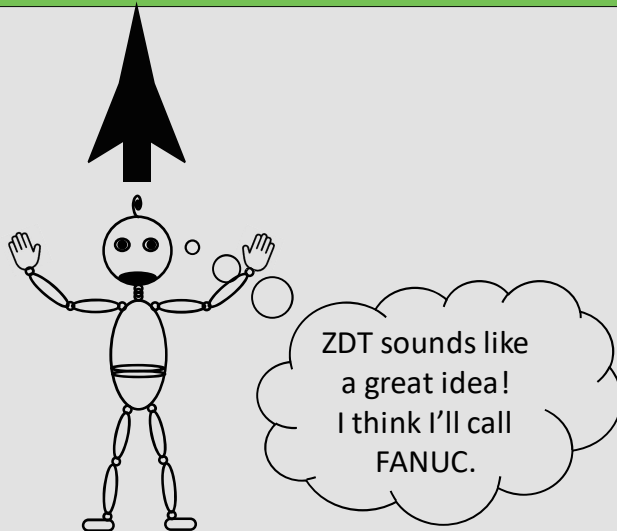
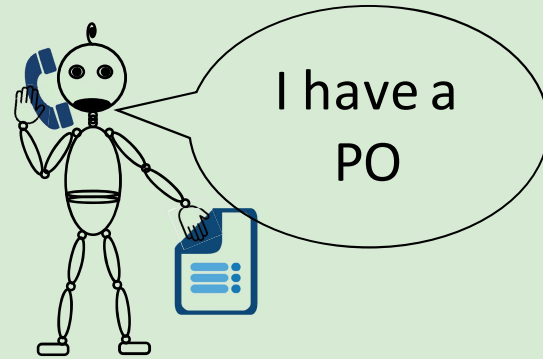


1. Portal Setup

Once you purchase ZDT, FANUC will create the portal through which you will monitor ZDT activities.

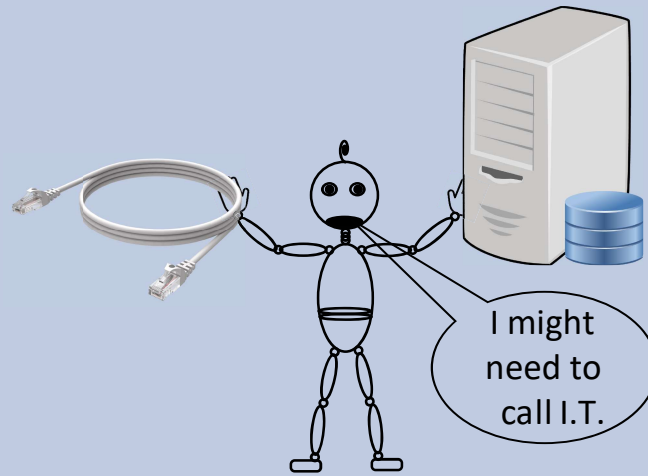
Once the portal is created, you can log in and set up your portal for your location.



2. Data Collector Configuration

The Data Collector collects and sends out data regarding robot health, maintenance, and connection status.

Connect the Data Collector to ensure a safe, reliable, and secure connection from the robot to FANUC.



4. ZDT Operations

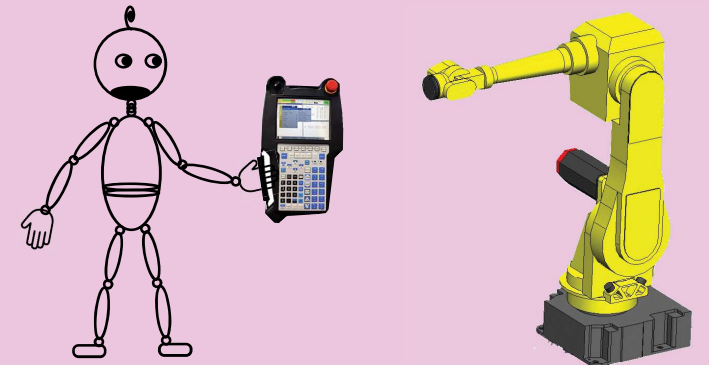
At this point, ZDT is active. Once you log in to the portal and assign any unassigned robots for monitoring, you can use the portal for monitoring, and take action on notification alert emails.

More information on using the ZDT web portal for these functions can be found in its online help system.



3. Robot Setup

Load any necessary robot options and enable ZDT functions. Program any necessary self-tests for ZDT reducer monitoring.



A. Log in to the Portal

1. When FANUC cRc receives the P.O. and ZDT Setup Sheet, they will create your ZDT Web Portal and send you an invitation similar to the one shown below.



2. When you get the invitation, click on **FANUC Robotics : Register** and fill out the registration form.

After agreeing to the listed policies, press the **Register** button.

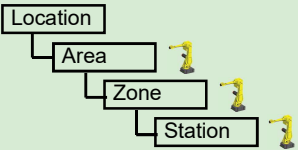
3. Once the registration is complete, you will be able to log into your ZDT Web Portal.

B. Create Your First Area

Note: Initially, your ZDT portal will be empty.

You must start to do some initial configuration, such as create your site's areas, zones, and stations.

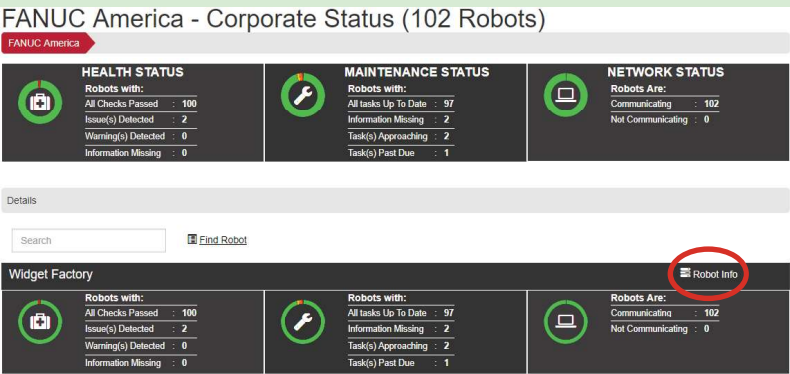
The ZDT portal organizes robots in a hierarchy.



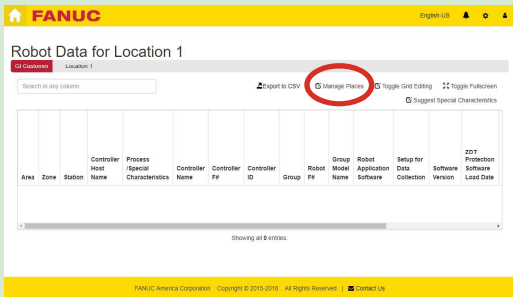
The Location is the highest level of organization in ZDT. When FANUC cRc created your web portal, they added your location for you. However, Locations cannot have robots directly assigned to them. Robots must be assigned to an Area, at the minimum.

Note: ZDT analysis does not begin until the robot(s) have been assigned to their area within the facility.

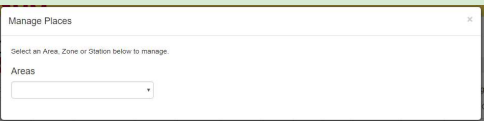
1. Click on the Robot Info link corresponding to your location.



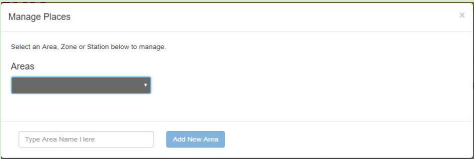
2. Click on **Manage Places** in the Robot Data grid.



3. Click on the **Areas** dropdown.



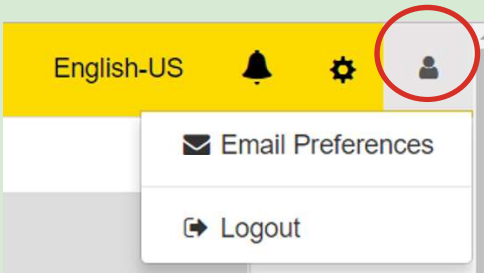
4. Type the name of your new area, then press the **Add New Area** button.



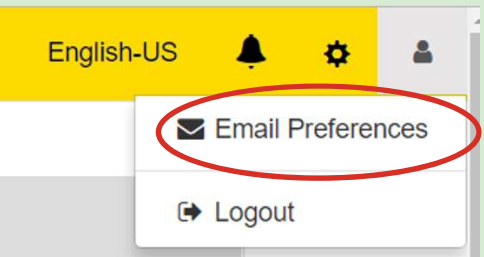
C. Set Email Preferences

The initial user in the portal is the company administrator. To set the email preferences for this user perform the following steps.

1. Click the User icon in the upper right.



2. Select **Email Preferences**.



3. For each location and area at your site, set the checkbox for the reports and notifications that the company administrator should receive.

These reports and notifications are defined as follows:

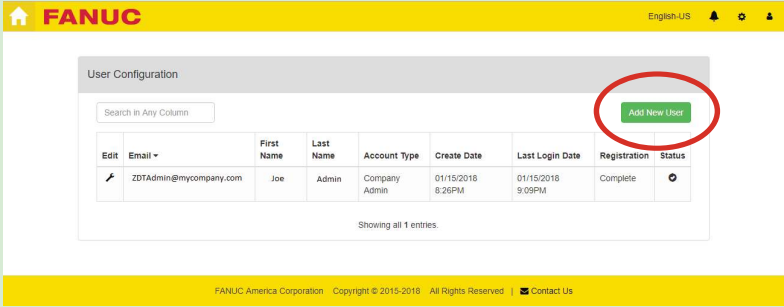
- **Comm Status:** Daily email used to track which robots at a location have been sending messages to the ZDT Data Center.
- **Insights:** Monthly email summarizing how robots at a location are functioning and where improvements can be made.
- **Alerts:** Automatic emails sent by analytics when issues are identified, that require immediate attention.
- **Daily Notifications:** Daily email summarizing how many alerts, warnings, and upcoming maintenance tasks have been detected at a location that day.

4. Press **Submit** to save the changes.

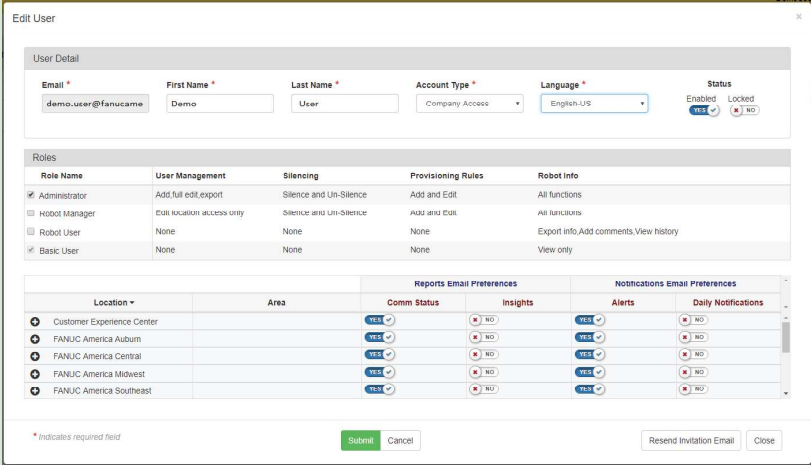
D. Add Other Users (Optional)

Note: The company admin user can add additional users and allow them access to the system.

1. Click the Gear icon in the upper right.
2. Choose **User Configuration**.
3. Click **Add New User**.



4. Fill in the Email (must be a company email address), First Name, and Last Name fields.

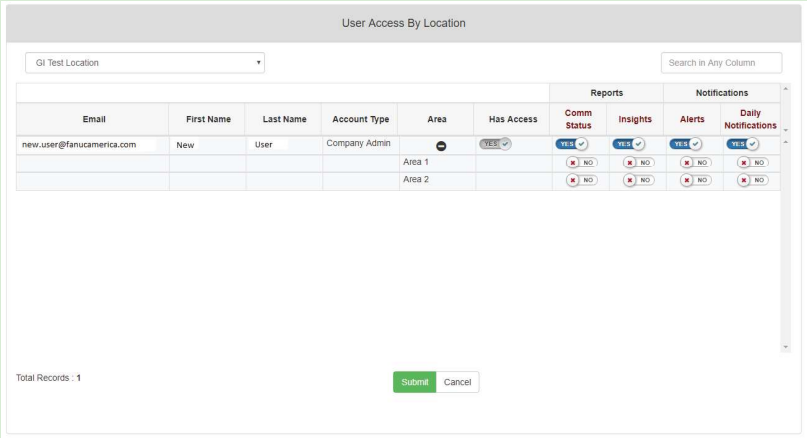


5. Set an appropriate Account Type:
 - **Company Access:** User is able to perform their assigned roles in all areas of all locations within the company.
 - **Location Access:** User is able to perform their assigned roles at all areas within specific locations in the company, as assigned by an administrator.
 - **Area Access:** User is able to perform their assigned roles only within specific areas at specific locations in the company, as assigned by an administrator.
6. Select the User's Role:
 - **Administrator:** Can perform all user management functions, can silence/un-silence warnings, can add/edit auto provisioning rules, can perform all robot editing and management functions.
 - **Robot Manager:** Can only grant/deny users access to locations, can silence/un-silence warnings, can add/edit auto provisioning rules, can perform all robot editing and management functions.
 - **Robot User:** Cannot perform any user management functions, cannot silence/un-silence warnings, cannot do anything with auto provisioning rules, can only add and view robot comments.
 - **Basic User:** Cannot perform any user management functions, cannot silence/un-silence warnings, cannot do anything with auto provisioning rules, can only view robot data.
7. Set the new user's default language.
8. Set the user's email notification preferences.
9. Press Submit.

E. Configure User Access (Optional)

Note: The company admin user can assign each user's access to different levels of the company hierarchy.

1. Click the Gear icon in the upper right.
2. Choose **User Access by Location**.

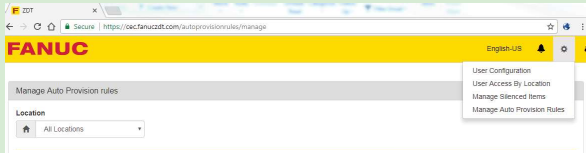


3. Find the user to be configured in the list.
4. Use the **Has Access** checkbox to allow or deny the user access to different hierarchy levels at a location.
5. Use the **Reports** and **Notifications** checkboxes to send or not send the corresponding reports and notifications to the user.
6. Press Submit.

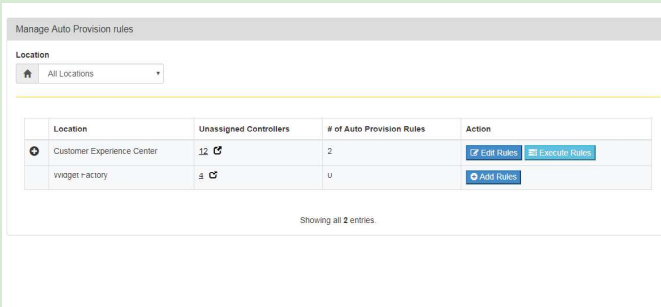
F. Rules for New Robot Connections (Optional)

To have your newly connected robots automatically assigned to their area of the facility, you can set up auto provision rules.

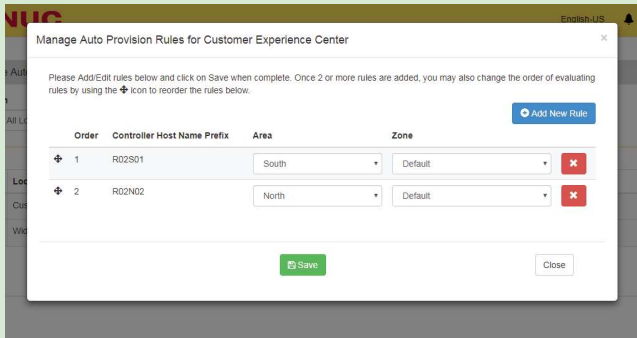
1. Click the Gear icon in the upper right.
2. Choose **Manage Auto Provision Rules** using the Manage Auto Provision Rules UI in the web portal.



3. From the dropdown menu, select your location and click on **Add Rules**.



4. Click on **Add New Rule**.



5. Enter your **Controller Host Name Prefix**.
6. Select the target area.
7. If you have a zone, select the target zone.
8. Press **Save**.

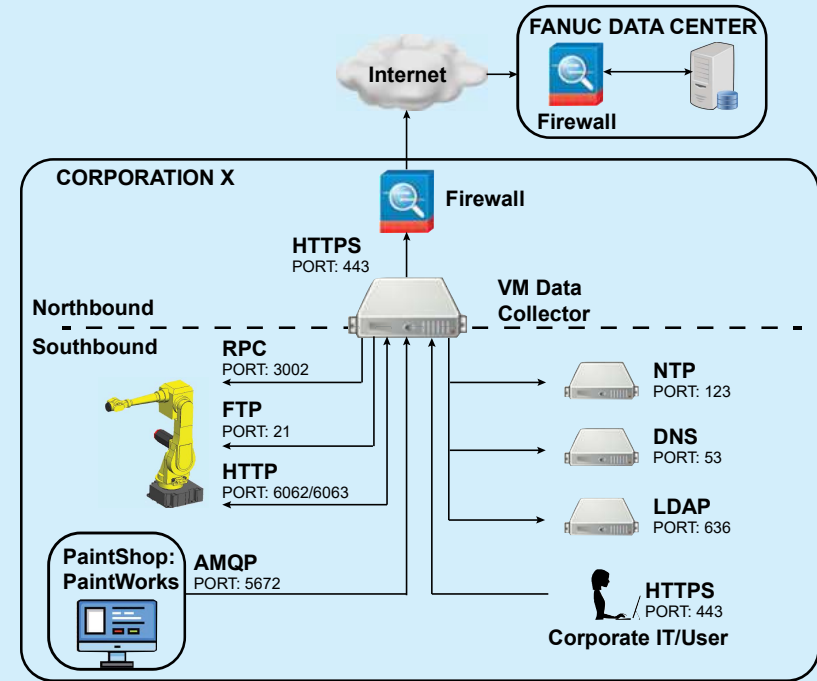
Portal setup is now complete.
Refer to the next section, VM Data Collector Startup.

Requirements

- You have experience setting up a virtual machine. If not, contact your IT Department/System Administrator for assistance.
- VMWare ESXi (6.5 or above) is installed on the server appliance.
- A server appliance for installing the VM, with the following at a minimum.
 - 6 CPUs
 - 8 GB RAM
 - 60 GB disk space
 - Network connection
- NTP Server
- One of the following browsers: Chrome (latest version) or IE (version 11 or above)
- OVA file from FANUC. **Note:** If you have not received this file, please contact FANUC at ZDTExecution@fanucamerica.com.
- ZDT provisioning file (.zxl) from FANUC. **Note:** If you have not received this file, please contact FANUC at ZDTExecution@fanucamerica.com.
- The **required** TCP network ports are open:

TCP Network Ports	Purpose
6062, 6063 and 5672	Robots send ZDT messages to ports 6062, 6063 and 5672 of the data collector.
3002	The data collector sends and receives data from robots using port 3002.
443	• The data collector sends secure messages to the cloud using URL fanuc01.wiprozdt.com on port 443. • The console UI is accessed with HTTPS on port 443 on the data collector.
21	Data sent and received via FTP.
LDAP port (typically 636)	(Optional) If LDAP is enabled in ZDT to allow additional users to log in to the console, the LDAP server port (typically 636) must be accessible from the ZDT Data Collector.

Important: Make sure that the ports shown above are open before proceeding.

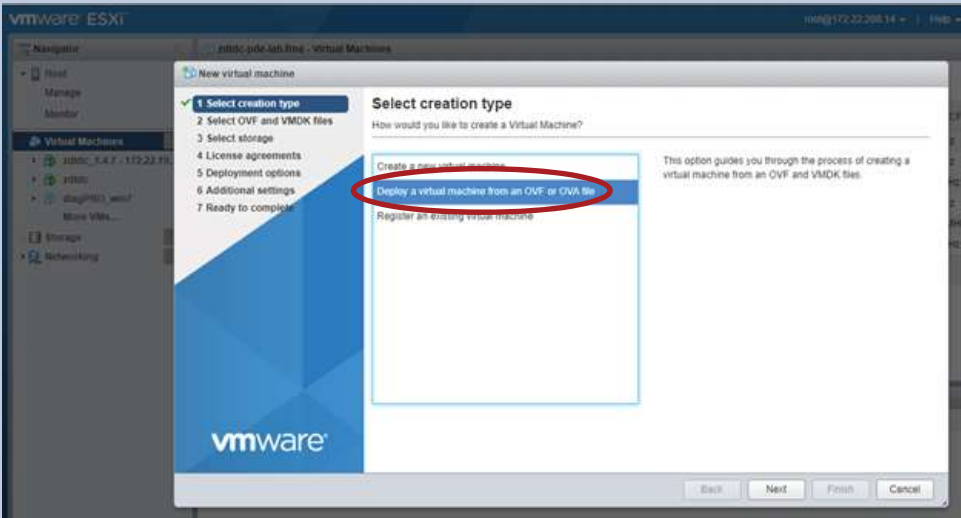


Additional Requirements

If you have an R-30iA controller, you must load a csv file on the data collector. This file is used to define the IP addresses of the robots. This file can be obtained by contacting FANUC at ZDTExecution@fanucamerica.com.

A. Creating the Virtual Machine (VM)

1. Contact FANUC at ZDTExecution@fanucamerica.com to get the OVA file, if not already done.
2. If you have a firewall and/or a captive portal, please open up the TCP ports shown in the table in the **Requirements** section.
3. The FANUC Data Collector is based on VMWare ESXi. Open your web browser, and log in to your VMWare ESXi host.
4. Click on **Virtual Machines**.
5. Choose **Create/Register VM**.
6. Select **Deploy a virtual machine from an OVA file**.

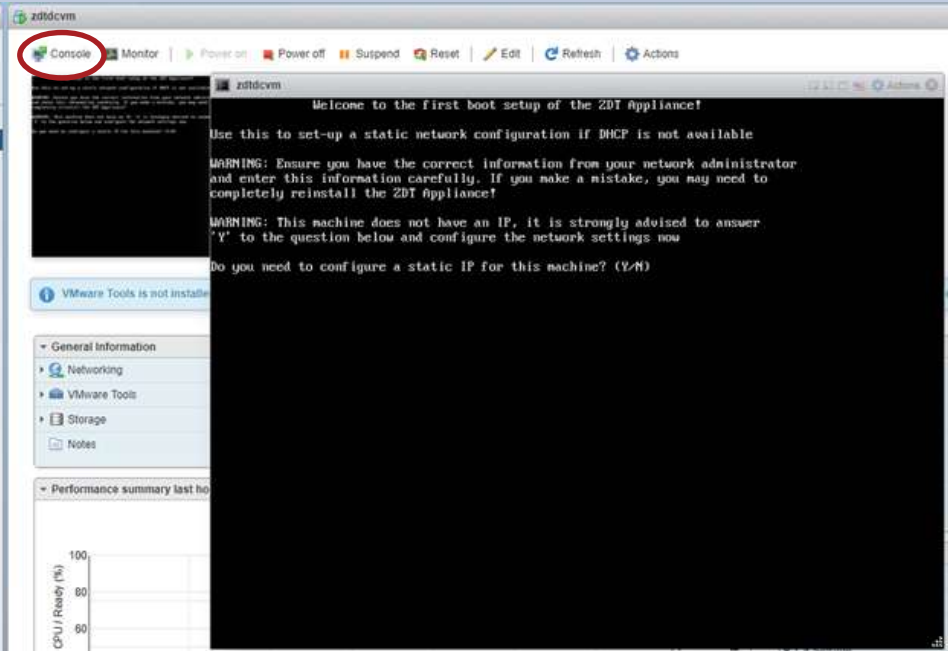


7. Press **Next**.
8. Drag and Drop the OVA file to the dialog box.

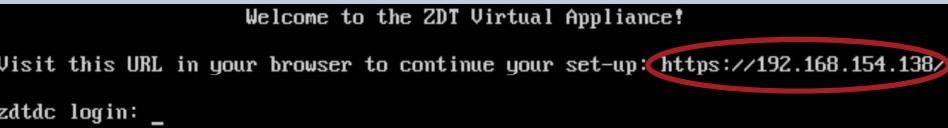
Note: Continue creating the VM. Contact your system administrator if you need assistance.

B. Configuring the Network

1. Select the Data Collector VM on ESXi, power it up, and then console into the VM to set up the network configuration for either DHCP or Static IP. A static IP is recommended.
- Note:** You won't be able to access the Data Collector VM and its web interface through the network later if you fail to provide the correct network settings.



2. The Data Collector IP address is displayed upon completion. Please remember this IP address. It is the ZDT Data Collector URL, and you will need it to browse to the ZDT Data Collector App in later steps. If you selected DHCP and do not see an IP address at this point, please check if there's a DHCP server running on the network.

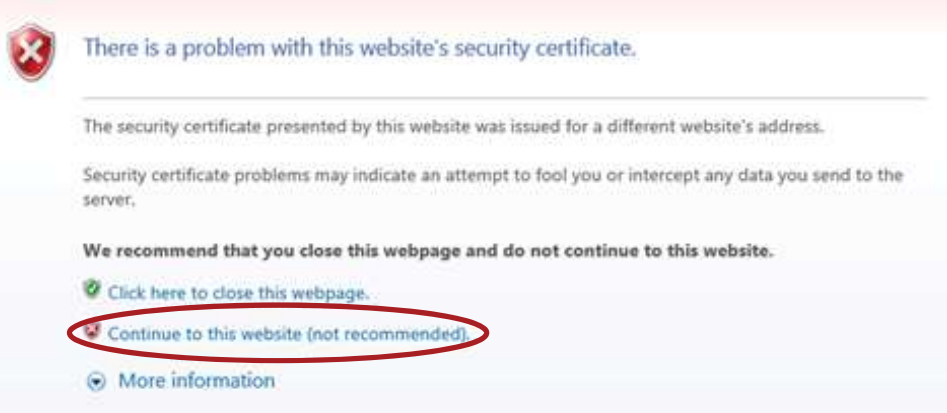


3. From the console, log in to the Data Collector VM using the username "zdtuser". The default password is provided by FANUC with the OVA file.
4. You will be prompted to change the zdtuser password. Follow the command line prompts to complete this step.
5. Type 'logout' to log out of the Data Collector VM after changing the default password.

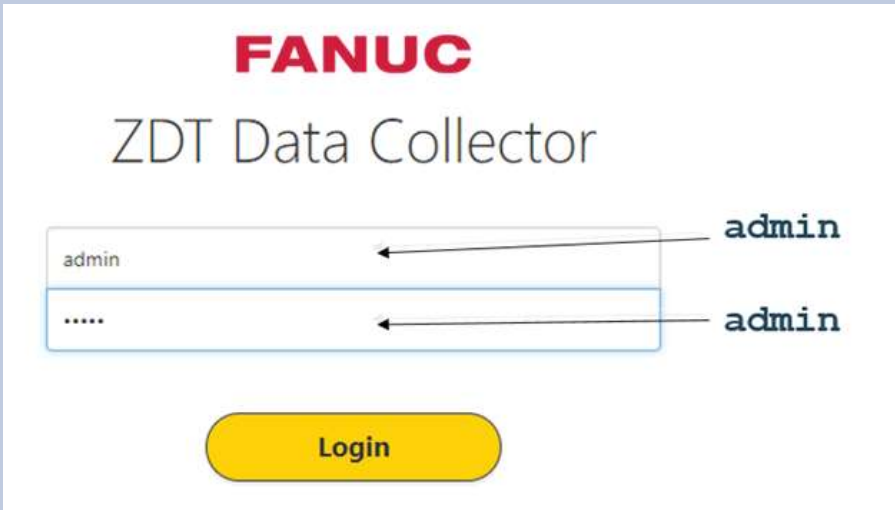
Note: From the time VM is up and running (step B1), it takes about two minutes for the ZDT Data Collector to startup and become available.

C. Log in to the Data Collector

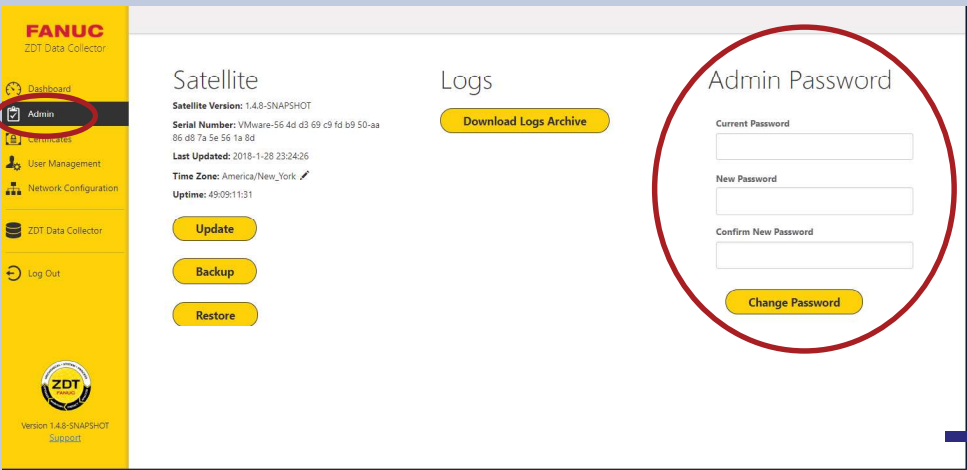
1. Open the web browser and enter the ZDT Data Collector URL that you captured in section B step 2 (e.g. https://192.168.1.2). Be sure to use the https prefix. You will see a warning screen similar to the one below.
- Note:** It may take a few minutes for the web interface to respond during the boot process.



2. Click **Continue to this website**. This will display the Data Collector login page.



3. Log in to the ZDT Data Collector with the following default username and password:
Username: admin
Password: admin
4. Change your administrator password.
- Click Admin.
 - Enter the current admin password.
 - Enter and re-enter a new password.
 - Click Change Password.



D. Provision your VM

Contact FANUC cRc (ZDTExecution@fanucamerica.com) to notify them that your device has been connected, to the internet to obtain the ZDT provisioning files. Note: You may have already provided this information via the ZDT Setup Sheet.

FANUC cRc will need the following information:

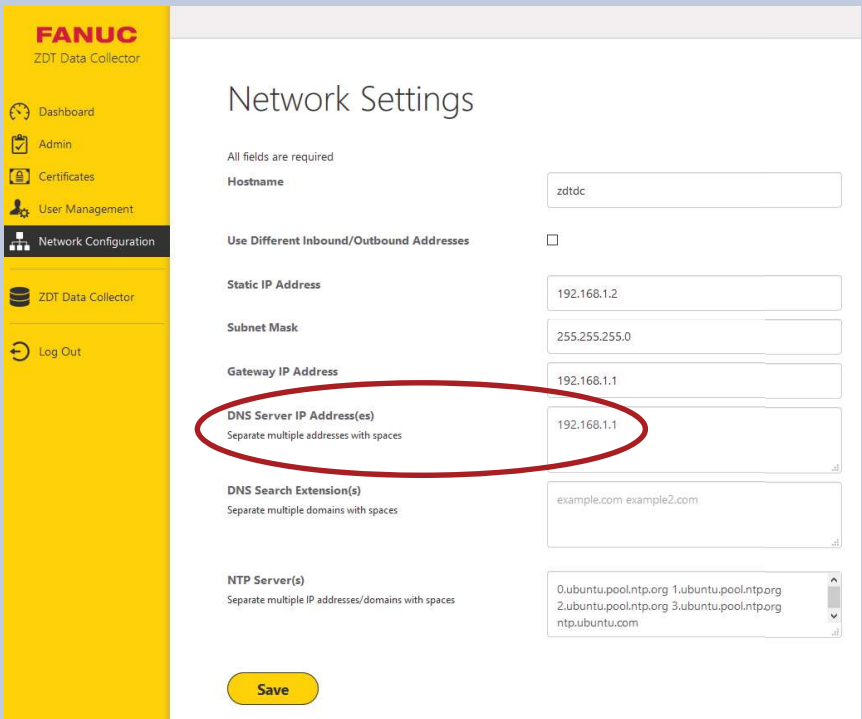
- IP Address of your VM
- ID of Robot Ethernet Port networked to the ZDT Data Collector (CD38A or CD38B found on Main CPU board)

FANUC cRc will provide you with each of the following files:

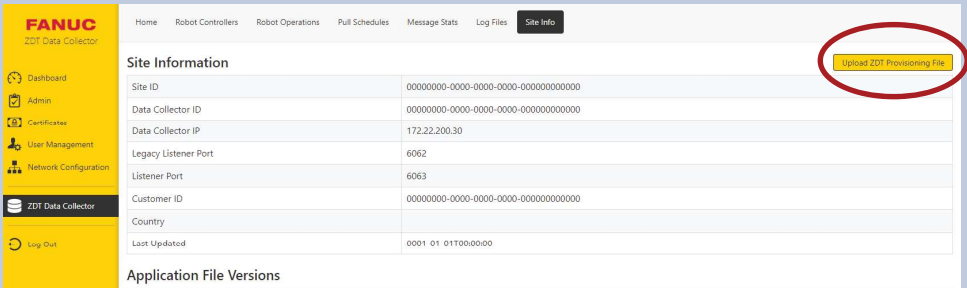
- **.ZXL** file which is used to configure the VM to send data to the ZDT Data Collector.
- **RCMCFG.XML** file which is used to configure your robots to communicate with your ZDT VM Data Collector. If the file you receive from FANUC has additional characters, please rename it to **RCMCFG.XML**.

1. Add the **DNS Server IP Address(es)** to Network Configuration.

Note: This step only applies if you are using a static IP address.

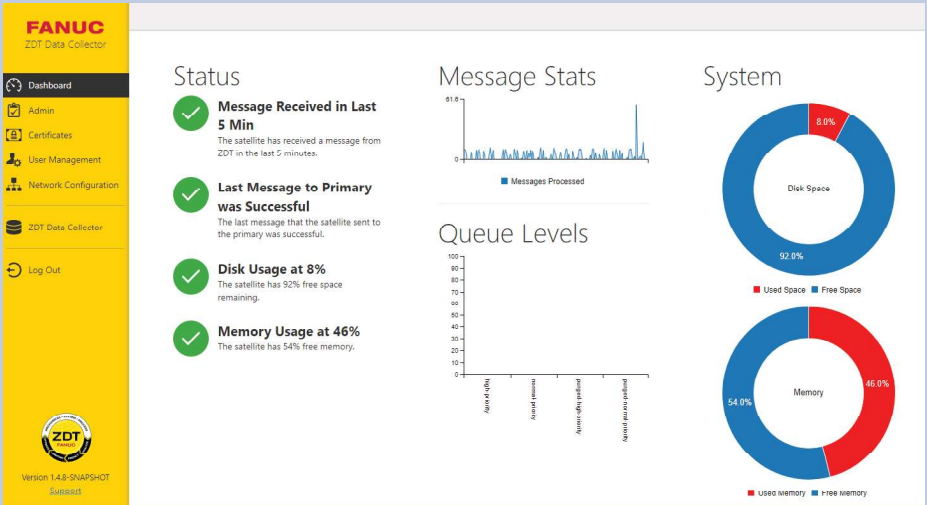


2. In the ZDT Data Collector screen, press **Upload ZDT Provisioning File** and load the .ZXL file provided by FANUC.



E. Verification

1. On the ZDT Data Collector Web UI, use the Data Collector Dashboard to check the status. Green indicates that the ZDT Data Collector is able to receive ZDT messages from the robot controller(s), and send them to the FANUC data center.



Status Descriptions

Message Received in Last 5 minutes: Green indicates that the Data Collector received messages from the robots within the last 5 minutes. Robots do not send messages to the Data Collector all of the time. When testing with one or more robots in idle, it is normal to see the status indicator turn green when you power on the robot(s) and then turn red after 5 minutes.

Note: The status indicator will remain red until you provision the Data Collector with the .ZXL site info file (required).

Last Message to Primary was Successful: Green indicates that the Data Collector is able to send the last message to the Primary. If red, see the following troubleshooting steps:

- **Incorrect network settings** - Verify that the configuration matches your network environment.
- **Network communication permission** – Contact your IT department to find out if permission needs to be granted for your new device to communicate over the network.
- **Incorrect proxy setting** - If using HTTPS proxy, verify the HTTPS proxy setting on the Data Collector or on the proxy server.
- **DNS failure** - Login to the VM, run: nslookup zdtcdprimary.fanuczdt.com. If the domain name zdtcdprimary.fanuczdt.com can't be resolved, please check the Network Configuration page and see if the DNS server address is correct. Also check that the DNS server is functioning properly.
- **HTTPS communication path to zdtcdprimary.fanuczdt.com or zdtcdprimary.fanuczdt.com can't be established** – If you have a whitelist (devices are only allowed to communicate to addresses on the list) only network, please contact your IT department to allow your Data Collector to communicate to zdtcdprimary.fanuczdt.com thru HTTPS port 443.
- **Satellite-To-Primary certificate has expired** - On the Certificates/Satellite-To-Primary page, check if the certificate has expired.

You are ready to configure your robots.
Refer to the next section, Robot Startup.

A. Robot Network Setup

Note: Your robot should already be connected to the same network as Data Collector.

1. From the robot Setup menu, choose **Host Comm** (MENU>>SETUP>>Host Comm).
2. Select **TCP/IP**.
3. After making sure you are on the correct port (Port#1 corresponds with CD38A and Port#2 corresponds with CD38B), set the following:
Note: The following IP addresses are examples only.
 - Robot IP: Use a unique and unused address in range, 192.168.1.2 - 192.168.1.128
 - Subnet Mask: 255.255.255.0
 - Router IP: 192.168.1.1 (If applicable)

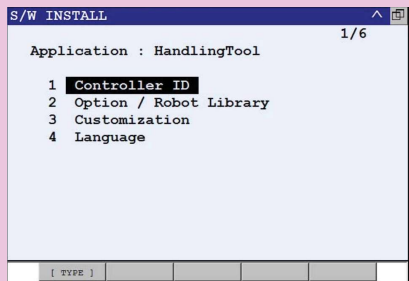
B. Install the ZDT Option

Note: If you have R-30iB Plus, you will not need to install the ZDT option. Please skip to, **C. Enable the ZDT Client**.

Note: If you have an R-30iB controller, software version 8.30p/32 or greater the following free option must be installed:

R882 (Application ZDT).

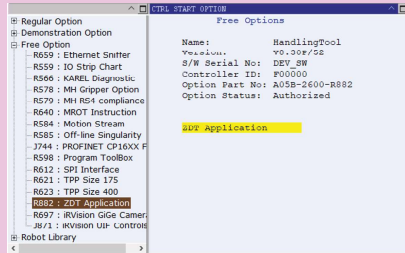
1. To load the option, enter Controlled start as follows:
 - a. From a powered down robot, turn the robot on while holding the **PREV** and **NEXT** keys.
 - b. From the Configuration Menu that appears, choose **Controlled start**. After a few minutes, the Controlled start screen will appear.
2. Insert original load media or the auto-update media into UD1.
3. Press **MENU**.
4. Cursor to and choose **S/W INSTALL**. You will see a screen similar to the following.



5. Choose **Option / Robot Library**.

B. Install the ZDT Option (continued)

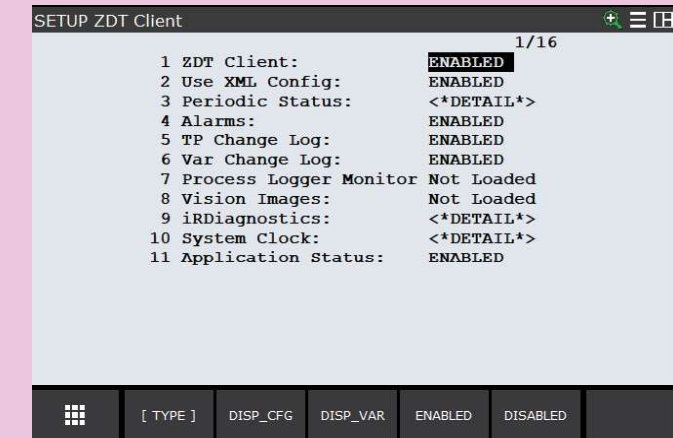
6. A screen will appear allowing you to choose the device that your software option is on. If it is not pointing to the proper device, press **F4, CHOICE** and select your device from the list.
7. Press **F3, DONE**. A screen similar to the following will appear.



8. In the left pane, open **Free Option** by pressing **+**.
9. Select **R882 (Application ZDT)** in the left pane by pressing **ENTER** and then press **F2, INSTALL**.
Note: Only Authorized options may be installed. If the option you wish to install is not Authorized, contact FANUC to get a Product Authorization Code (PAC). Using this PAC code, you can Authorize and then Install the option.
10. To get out of Controlled start, press **FCTN** and choose **START (COLD)**.

C. Enable the ZDT Client

1. Press **MENU>>SETUP>>ZDT Client** to go to ZDT Client Setup Screen.
2. Select **ZDT Client** and change to **ENABLED**.
3. Select **Use XML Config** and change to **ENABLED**.



4. Make sure the **System Clock** is set properly.
 - a. Cursor to the **System Clock** and press **ENTER**.
 - b. Press the **DISP** key to enter the right pane.
 - c. Press **F4, ADJUST**.
 - d. Cursor to the appropriate field and make your changes.
 - e. When finished, press **F4, FINISH**.

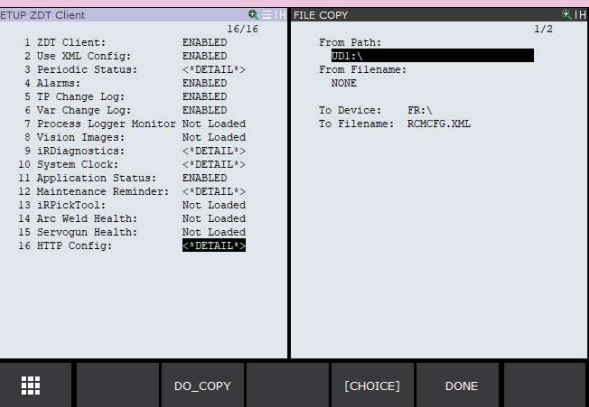
D. Load the RCMCFG.XML File

Note: The RCMCFG.XML file defines the IP address of the ZDT Data Collector where ZDT data is sent. If you have not received this file yet, contact ZDTEExecution@fanucamerica.com.

1. After receiving the file from FANUC, rename it **RCMCFG.XML**.
2. Put the RCMCFG.XML file in the root directory of a USB flash drive and insert the USB flash drive into the controller's USB Port (UD1 or UT1).
3. Press **MENU>> SETUP>>ZDT Client** to go to the ZDT Client Setup Screen.
4. Cursor to **HTTP Config**: and press **ENTER**. You should see a screen similar to the following.



5. With the focus in the right pane, select **F2, IMPORT**.
6. Choose **F4, RCMCFG**. You should see the following screen.

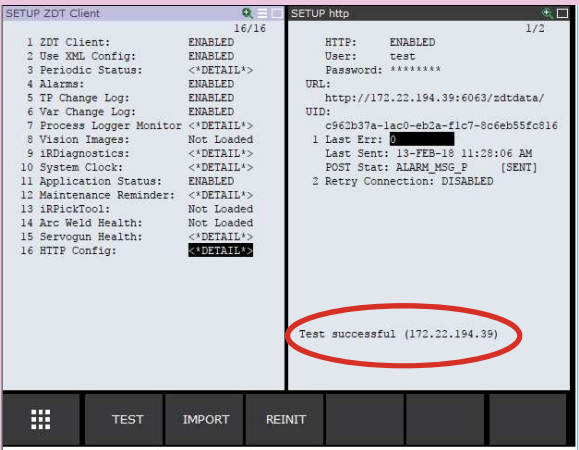


7. Select **From Path**, press **F4, CHOICE**.
8. Choose the device where RCMCFG.XML is stored.
9. Select **From Filename**, press **F4, CHOICE**.
10. Choose **RCMCFG.XML**.
11. Select **F2, DOCOPY**. If prompted to overwrite, press **F4, YES**. The screen will show **Copied to FR:\RCMCFG.XML**.
12. Select **F5, DONE**.
13. Select **F3, REINIT**.

E. Validate Installation of the RCMCFG.XML file

1. Press **MENU>>SETUP>>ZDT Client** to go to the ZDT Client Setup Screen.
2. Select **HTTP Config**: and press **ENTER**.
3. Validate that the Data Collector IP Address matches the plant's data collector IP address.
4. The server can be validated using the **F1, TEST** button on the HTTP configuration screen. One of three scenarios is possible.
 - The Data Collector host is not reachable. This will result in the message "Ping: invalid Host".
 - The server host is live, but the web service is not running. The message will read, "host OK and service NOK (zdtserver.com)" where zdtserver.com is the host name of the server in <http://zdtserver.com:6063/zdtdata/>.
 - When service and the server host are up, "Test Successful (zdtserver.com)" is reported. The example below shows 172.22.194.18 instead of zdtserver.com.

5. If the test is successful, you should see the following screen.



Note: Your robot(s) should be communicating. Allow about 24 hours for your robot and its data to show up in the web portal.

F. Enable Features on the ZDT Client Menu

1. A setup screen is provided to enable/disable the sending of ZDT data to the Data Collector. To access the Setup ZDT Client screen: **MENU >> SETUP >> ZDT Client**.
2. Press **i** + **FCTN** to display details of each selection. Under **Related Views**, choose **ZDT Client Details**.
3. Unless informed by FANUC, please enable all ZDT features. that have not already been enabled. Some features, when enabled in the left pane, have associated items in the right pane that also need to be enabled. Your screen will reflect the options loaded at your facility. Make sure that you have enabled **iRDiagnostics** and any sub-menus.



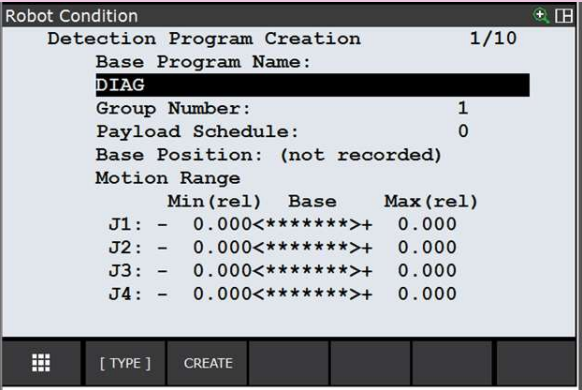
4. Cycle robot power.

Note: Not all robots support the reducer and motor health. Contact FANUC to see if your robot is a candidate. If your robot does not, you can skip section G and H.

G. Reducer Health: Condition Program Setup

Note: The Reducer Health feature requires the execution of a Robot Condition or self-test program to be executed on a regular basis.

1. Before you begin, verify that your robot is mastered and calibrated.
2. Navigate to the robot condition setup menu: **MENU>>UTILITIES>>Robot Condition**.



3. Move the cursor to the **Base Program Name** and provide a name for the diagnostic program, e.g. **DIAG**.
4. Move the cursor to **Group Number**, and type the number of the group for analysis.
5. Establish a base/start location for the self-test.
 - a. Jog the robot to a safe position that facilitates collision free motion (within a limited range).
 - b. Record the current robot position as the **Base Position**. You must be in T1/T2 or Teach Mode. Press and hold **SHIFT** and **F4, RECORD** to record the **Base Position**.
6. Set the Payload Schedule to the correct payload.
7. Enter an allowable motion range for each axis in degrees. For each axis, enter the allowable motion range relative to the base position, both MIN and MAX. Normally this is +/- 10 degrees. When the test is run, the robot will move one axis at a time.

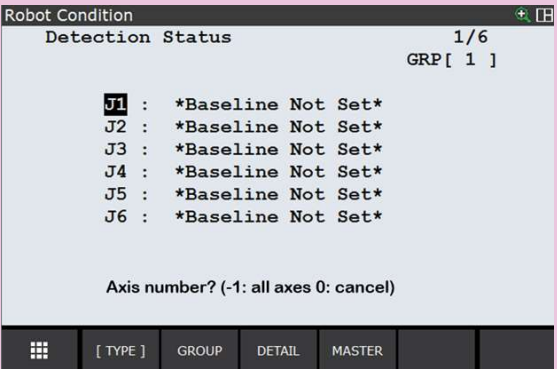
Note: Aux or extended (rail) axes are not included in the self-test.

8. Disable the teach pendant and switch to T2 or AUTO mode (paint systems would be Manual or Production mode) and reset all faults.
9. Set the override to 100%. Press **SHIFT** and **F2, CREATE**. A program will be created with a group number added to the Base Program Name.

Note: It is normal for the robot to fault when the program creation is complete.

G. Reducer Health: Condition Program Setup (continued)

10. Run the program just created, in this case **DIAG1**, with a low override to assure that the robot clears all obstructions.
11. Run the program five times at 100% to warm up the robot for more accurate results.
12. Navigate to the menu to establish the baseline data: **MENU>>NEXT>>STATUS>>Robot Condition**.
13. Press **MASTER**. The prompt, **Axis number? (-1: all axes 0: cancel)** will appear as shown below.



14. If this is the first time the Robot Condition Baseline has been mastered, enter **-1**. If you want to master a single axis, perhaps after some maintenance has been performed, enter the desired axis number. If you wish to cancel Robot Condition Mastering, enter **0**.
15. Contact the controls engineering group of your facility and request that the generated condition program be executed at least once a day. This may require PLC changes, etc.
16. Remember to have the Condition Program run with the same conditions each time. For example: if it was generated without a part, always run the program without a part.

H. Motor Health: High Water Mark Job Data Setup (No setup needed if you do not have looping programs)

Note: High Water Mark Job Data monitors the TP program running status. No extra setup is required unless your TP programs(s) are looping. If you do not have looping programs, please skip this section.

1. Navigate to the **iRDiagnostics** section of the ZDT Client Screen: **MENU>>Setup>>ZDT Client>>iRDiagnostics>>Enter**.
2. Press **DISP** to focus on the right screen, cursor to **5 Mode**: and press **[CHOICE]**.



You will see five Job Data Modes. We recommend using one of the following:

- **Auto(sub-progs)**: This is for continuously looping programs. Selecting this mode, HWM Job Data is recorded when any sub TP program (CALLED program) with motion lines is called. Data is sent when the sub TP program returns back to the main program.
- **I/O Trigger**: An I/O point is used to trigger the Job Data. I/O type, port number, and trigger type (when the I/O turns on or off) can be set by using the menu items shown below:



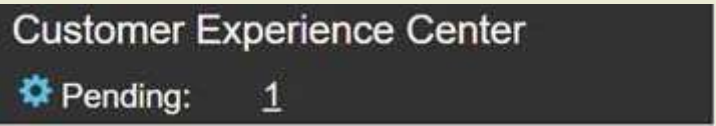
Robot setup is now complete.
Refer to the next section, Operations Startup.

A. Assign Any Unassigned Robots to Areas

Now that your robots are connected to ZDT, your robots will begin sending data to the ZDT cloud.

Note: It may take 30 minutes for your robots to show up in the portal. If you do not see your robots online in that time, please log on to the Data Collector UI to check the status.

Your robots will remain "Pending" until the ZDT system has processed the robot subscription. This could take up to 1 day. When the robot is out of "Pending" status, analytics will begin to run.



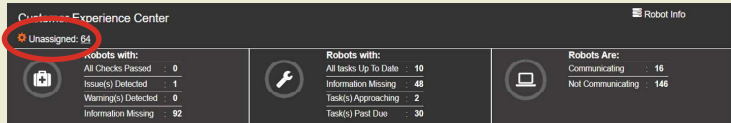
If you do not have auto provision rules, or you have robots that are online but do not match the rule pattern, they will be "Unassigned." They will be called out in a banner similar to that shown below, at the top of the portal page.



Analytics will NOT run on Unassigned robots. To have analytics run on their data, you must manually assign them to an area.

To assign a robot to an Area, follow the steps below.

1. Click on the **Unassigned** link for your Location.



2. Click on **Toggle Grid Editing** in the Robot Data grid.

Robot Info for Customer Experience Center											
Search in any column											
FAQ: How to Change Your F-Number											
Area	Zone	Station	Controller IP	Controller Host Name	Process / Special Characteristics For All Groups	Controller Name	Controller F#	Controller ID	Group	Robot F#	Robot Application Software Version
South	Arc Welding Cell	Arc Weld	172.22.66.88	ARCROB14	Sends_Alarm_Msg_T, Sends_batt_low2_T, Sends_rc_1	ARCROB14	164817	c962b37a-1a0d-4b2a-81a7-80a8a13b06c	Group 2	164817	LR ArcTool 200ED/TL V8.30P/32
South	Arc Welding Cell	Arc Weld	172.22.66.00	ARCROB14	Sends_Alarm_Msg_T, Sends_batt_low2_T, Sends_rc_1	ARCROB14	164817	530b07a-1a0d-4b2a-81a7-80a8a13b06c	Group 1	164817	M-100A/12 LR ArcTool V8.30P/32

3. Click on the **Unassigned** link under Area, then select the desired area from the drop-down.

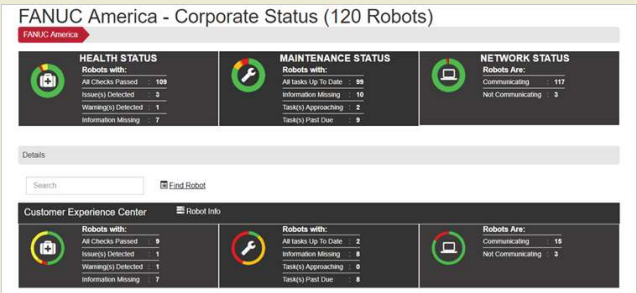
Note:: If there are no areas listed here, please refer to the Portal Startup Guide to create your areas.

Location		Controller IP	Controller Host Name	Process /Special Characteristics
Test Location	Select Area ▼ Select Area Area 1 Unassigned Unassigned save cancel	1.2.3.4	New Robot	empty

4. Press the **save** button.

B. Navigate/Use the Web Portal

1. Examine the company level dashboard.



The ZDT portal shows status in three main areas: Health, Maintenance, and Network.



Health Status Panel
Indicates the current results of analytics categorized as: Mechanical Health, System Health, Process Health



Maintenance Status Panel
Indicates the current status of scheduled maintenance items

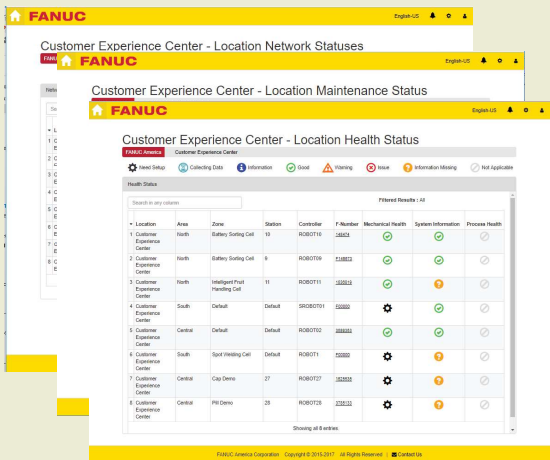


ZDT Network Status Panel
Indicates the current status of robots sending messages to the ZDT data center

- Each of these areas has its own panel with a circular gauge that displays the current status using different colors.
 - Green:** How many robots have no issues.
 - Red:** How many robots require immediate attention.
 - Orange and Yellow:** How many robots have warnings that should be investigated.
- The company level dashboard shows the overall company status bar at the top.
- The individual locations within the company have their own bars in the details section below the company bar.
- Each gauge, the numbers, and the panels can be clicked to access screens with more information about the statuses.
- Areas and zones defined in the company hierarchy will have their own dashboards with similar status bars.
- The **Find Robot** tool allows you to find your robot by serial number, controller identification number, station or zone.

2. Examine the Health, Maintenance, and Network status screens.

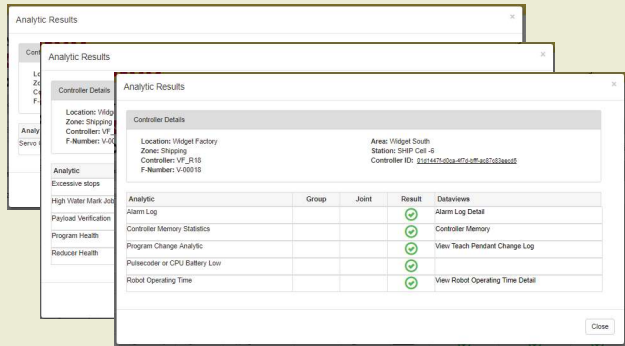
- When a Health, Maintenance, or Network status indicator is clicked from the dashboard, a detailed listing of robot controllers and their status is shown.



- Clicking on the status icon for a robot controller will open up a detailed status pop-up.
- Clicking on a robot's F-Number will open that robot's Home Page.
- Typing text in the search box will filter the list of robots.

B. Navigate/Use the web Portal (continued)

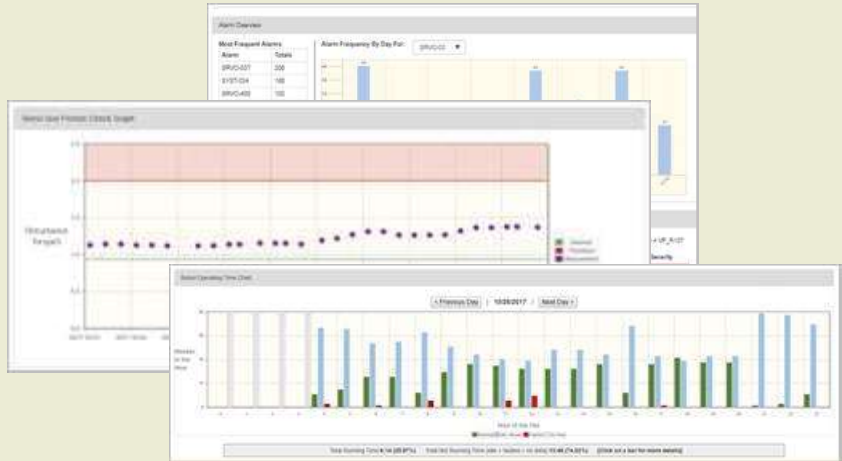
3. Examine the controller detailed status pop-ups.



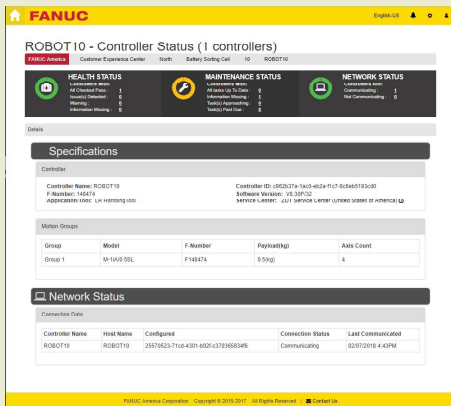
- When a controller Mechanical, System, or Process Health status icon is clicked, a pop-up with individual analytic results is displayed.
- Each analytic is listed with an icon showing the most recent result, and a link to the corresponding data view.
- If an analytic is configured for a specific group and/or joint, that information is also displayed.
- Some analytics do not have data views, so there will be no data view link.
- To access an analytic data view, click on the link next to the status icon.

4. Click on **Dataviews** link to access Analytic dataviews.

- Analytic data views contain information about what the analytic found when it examined a robot's data.
- Analytic data views can be accessed from a link in the status pop-up in the portal or from a link in an alert email.
- Each analytic data view will have its own specific UI characteristics and operation.



5. Examine the robot home page.



- The robot home page contains information about a specific robot controller.
- The robot home page can be accessed from various portal pages by clicking on F-Number or controller ID links.



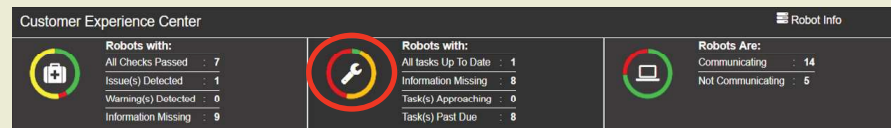
C. Set Maintenance Dates

Note: You will be unable to Set Maintenance Dates until the daily analytic config job is finished, and the robot is no longer in *pending* status.

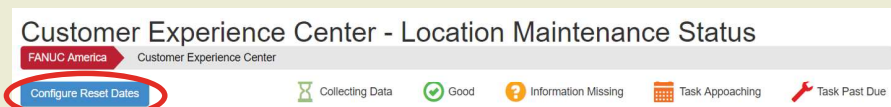
ZDT will need to know when you last performed scheduled maintenance on your robots so it can accurately remind you when the next maintenance is due.

To set maintenance dates, follow the steps below.

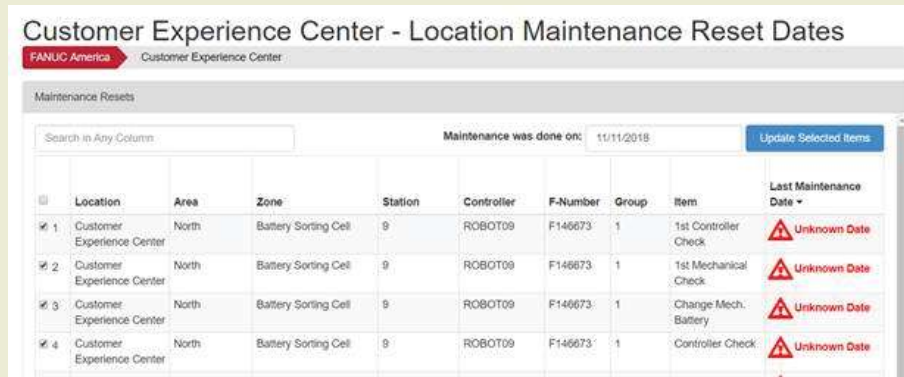
1. Click on the wrench icon in the Maintenance panel.



2. Click on the **Configure Reset Dates** button.



3. Enter the maintenance date in the **Maintenance was done on** text-box.



4. Check the box next to each robot; each maintenance item that was done on that date.

5. Press the **Update Selected Items** button.

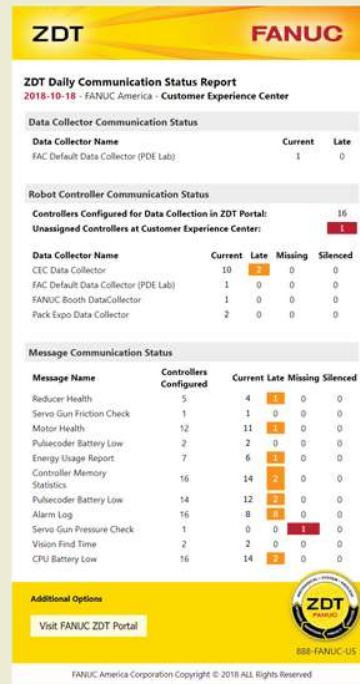
6. Repeat steps 3-5 for each date on which maintenance was done.

D. Take Action on Reports and Notifications

1. Review Daily Communication Status Report.

Every day a Communication Status email is sent that summarizes the current state of data collector and robot communications. The status of each device is broken down as follows:

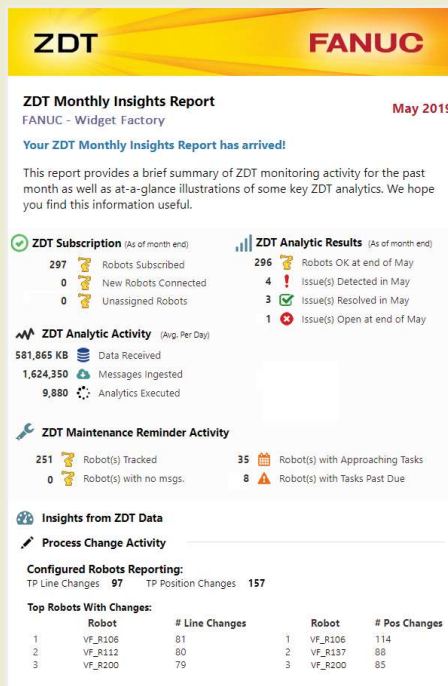
- Current: Data has been received in the last 24 hours.
- Late: Data has not been received in the last 48 hours.
- Missing: Data has never been received.



2. Review Monthly Insights Report.

This includes summaries of:

- Subscription and Analytic summary
- Maintenance summary
- Robot utilization
- E-Stop information
- Energy information

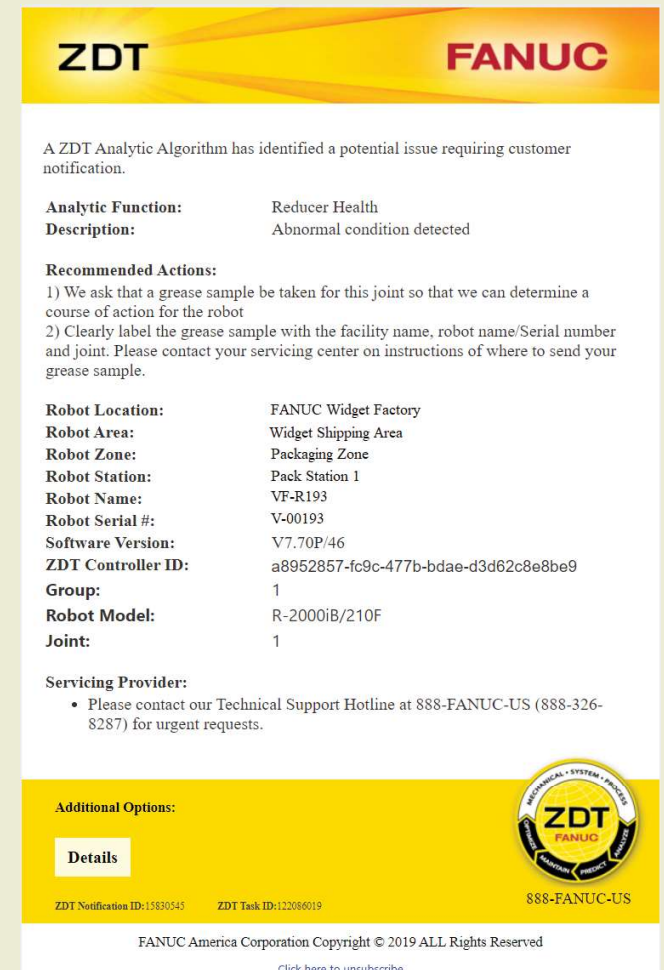


E. Take Action on Alerts

When a ZDT analytic detects a condition that requires immediate attention, you will receive an alert email. The email contains a description of the function along with recommended actions. There may also be official documentation you can download. The specific robot with the alert will be listed, along with its Area, Zone, and Station.

1. Clicking on the Details button at the bottom of the email will take you to the corresponding analytic data view on the ZDT portal where more information can be found.

FANUC cRc will also be notified of the alert and will be available to provide assistance. Once the issue is found, please reply to the Alert email with any action taken.



Note that the notifications icon on the ZDT Web Portal header provides access to all active Issues, Warnings and Maintenance.



For more information

To access controller, robot, or application-specific manuals, scan the QR Code to the right or go to <https://www.fanucamerica.com/qr/edocs>.

United States: 1-888-326-8287

Canada: 1-905-812-2300

Mexico: + 52 (449) 922-8000

South America: + 55 (11) 3619-0599

Or, select your region at <https://www.fanuc.com>.

